第四讲 海水运动方程（2）

第 5 章 海水基本运动方程 （5.1, 5.10 ~ 5.11）

- ◆ 海洋中的力
- ◆ 全微分的物理意义
- ◆ 动量方程的导出

Introduction to physical oceanography（R. H. Stewart)，2008. Chapter 7

Science 公布的125 个科学前沿问题

23.温室效应会使得地球温度达到多高？

43.能否发展关于湍流动力学和颗粒材料运动学的综合理论？

122.数学家将会最终给出Navier-Stokes方程的解吗？

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23 WHAT DON'T WE KNOW?

Special Section

How Hot Will The Greenhouse World Be

varied naturally in the distant past and how the climate system of the time responded. Nature, of course, has never run the perfect analog for the coming greenhouse warming. And estimating how much carbon dioxide concentrations fell during the depths of the last ice age or how much sunlight debris from the eruption of Mount Pinatubo in the Philippines blocked will always have lingering uncertainties. But paleoclimate estimates of climate sensitivity

43 Can we develop a general theory of the dynamics of turbulent flows and the motion of granular materials?

122 Will mathematicians unleash the power of the Navier-Stokes equations?

Will mathematicians unleash the power of the Navier-Stokes equations?

数学家们能否真正释放纳维 – 斯托克斯方程的强大威力？

First written down in the 1840s, the equations hold the keys to understanding both smooth and turbulent flow. To harness them, though, theorists must find out exactly when they work and under what conditions they break down.

蕴含着理解层流和湍流的关键

运用这些方程组，理论学家必须准确弄清楚

它们在什么条件下适用，以及在什么条件下会失效

克洛德·路易·马里·亨利·纳维 是一位法国工程师与物理学家，主要贡献在力学领域。著名的纳维 - 斯托克斯方程以他和乔治·加布里埃尔·斯托克斯命名。

乔治·加布里埃尔·斯托克斯爵士，**第一代从男爵**，FRS，爱尔兰数学家和物理学家，就读和任教于剑桥大学，主要贡献在流体动力学（如纳维-斯托克斯方程）、光学和数学物理学（如斯托克斯公式）。他曾任皇家学会秘书和会长。

海洋中的力 (5.1)

重力(gravity)	引潮力 浮力 水平压强梯度
摩擦力(friction)	风应力 底部切应力
惯性力(inertial force)	表观力 伪力

全微分

$$q_{out} = q_{in} + \frac{\partial q}{\partial t}\,\delta t + \frac{\partial q}{\partial x}\,\delta x \tag{7.5}$$

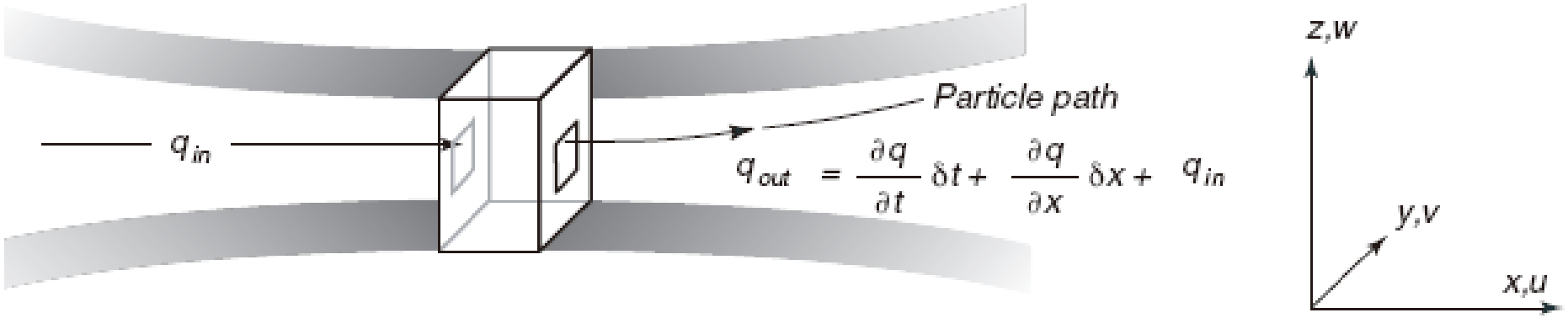


Figure 7.2 Sketch of flow used for deriving the total derivative.

The rate of change of the quantity q within the volume is:

$$\frac{Dq}{Dt} = \frac{q_{out} - q_{in}}{\delta t} = \frac{\partial q}{\partial t} + \frac{\partial q}{\partial x}\frac{\delta x}{\delta t} \tag{7.6}$$

But $\delta x/\delta t$ is the velocity u, and therefore:

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + u\frac{\partial q}{\partial x}$$

In three dimensions, the total derivative becomes:

$$\frac{D}{Dt} = \frac{\partial}{dt} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} \tag{7.7a}$$

$$\frac{D}{Dt} = \frac{\partial}{dt} + \mathbf{u} \cdot \nabla(\,) \tag{7.7b}$$

where $\mathbf{u}$ is the vector velocity and ∇ is the operator *del* of vector field theory (See Feynman, Leighton, and Sands 1964: 2–6).

This is an amazing result. Transforming coordinates from one following a particle to one fixed in space converts a simple linear derivative into a non-linear partial derivative. Now let's use the equation to calculate the change of momentum of a parcel of fluid.

这是个惊人的结论。从跟随质点运动的坐标系（拉格朗日坐标系）变换到固定在空间中的坐标系（欧拉坐标系），将一个简单的线性全导数（质点导数）转化为非线性偏导数。

运动方程

- 牛顿第二运动定律

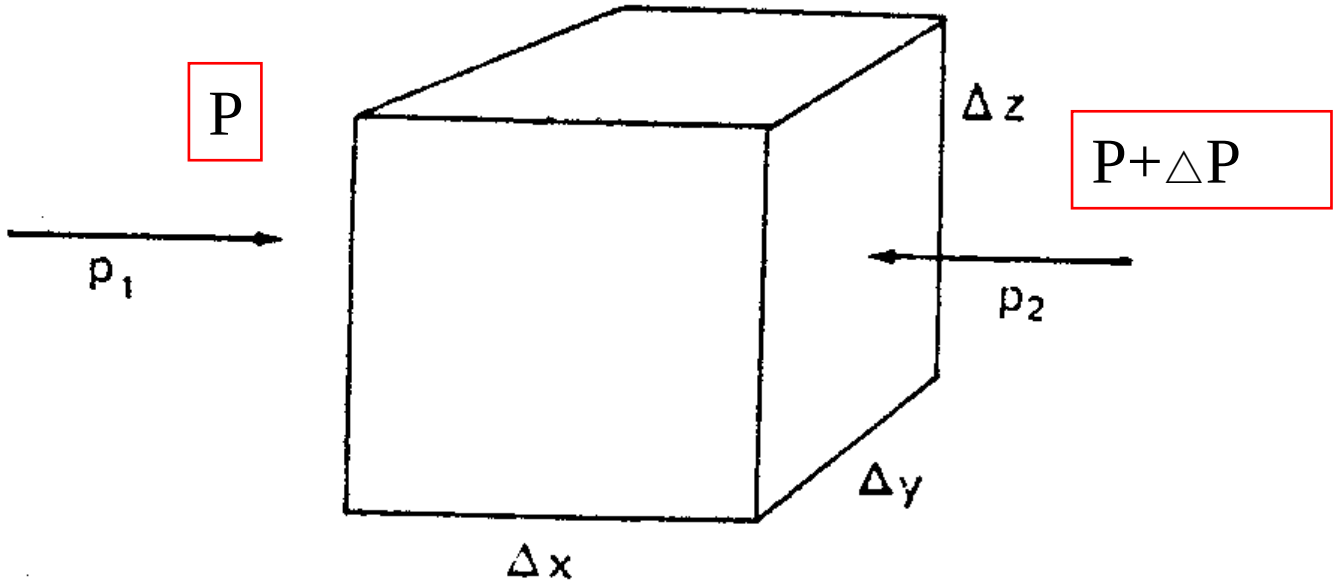
$$F = m\,a$$

运动方程

- ◆压强梯度力
- ◆重力
- ◆科氏力
- ◆Navier-Stokes方程、Euler 方程
- ◆边界条件
- ◆静压假定

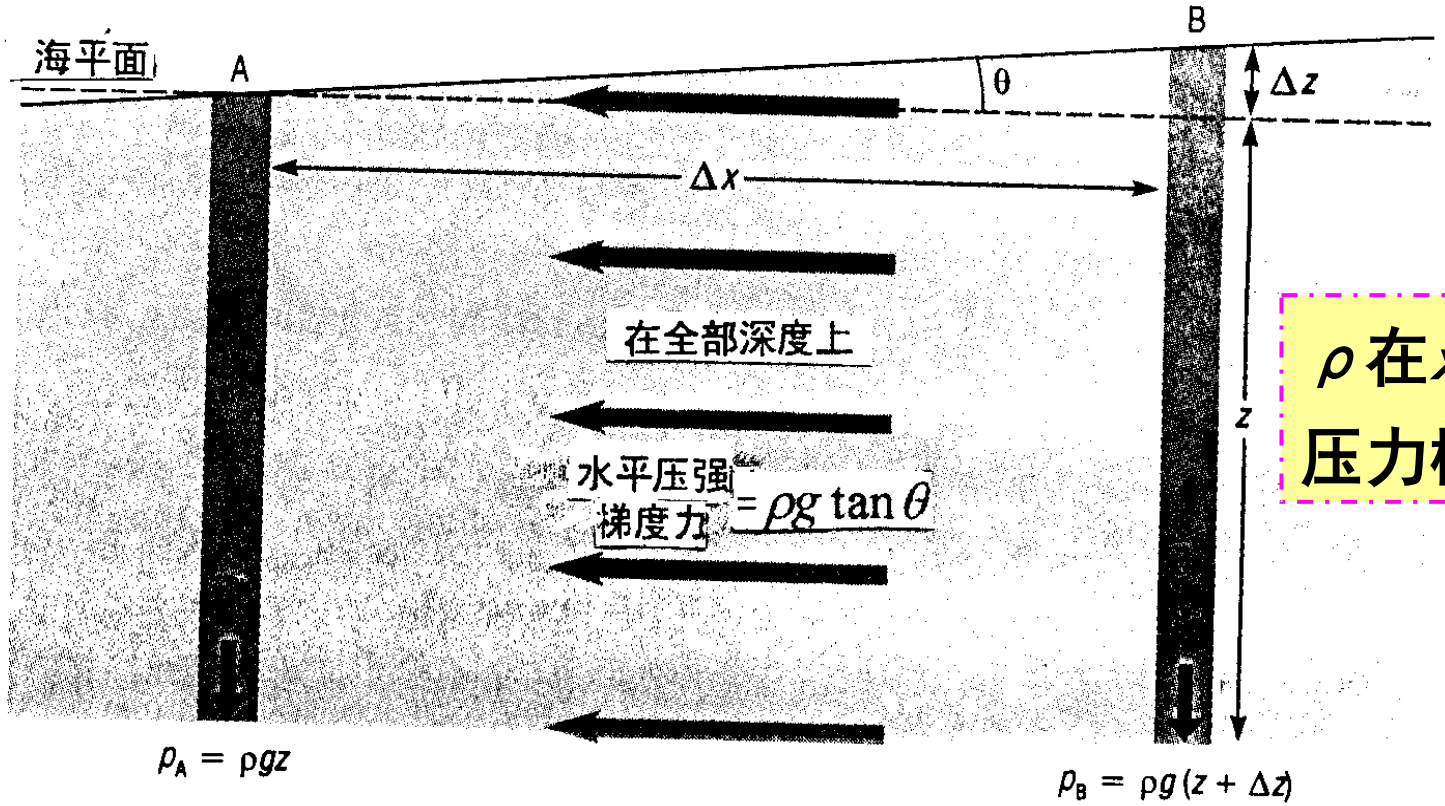
压力梯度力

定义：由于流场中压力分布不均匀而产生的，推动流体质点自高压向低压方向运动的力



- ◆立方体的流体质点
- ◆背景压力场沿x方向增加 p2>p1

压力梯度力



ρ 在 x 和 z 方向上不变，
压力梯度力在任何位置都相同

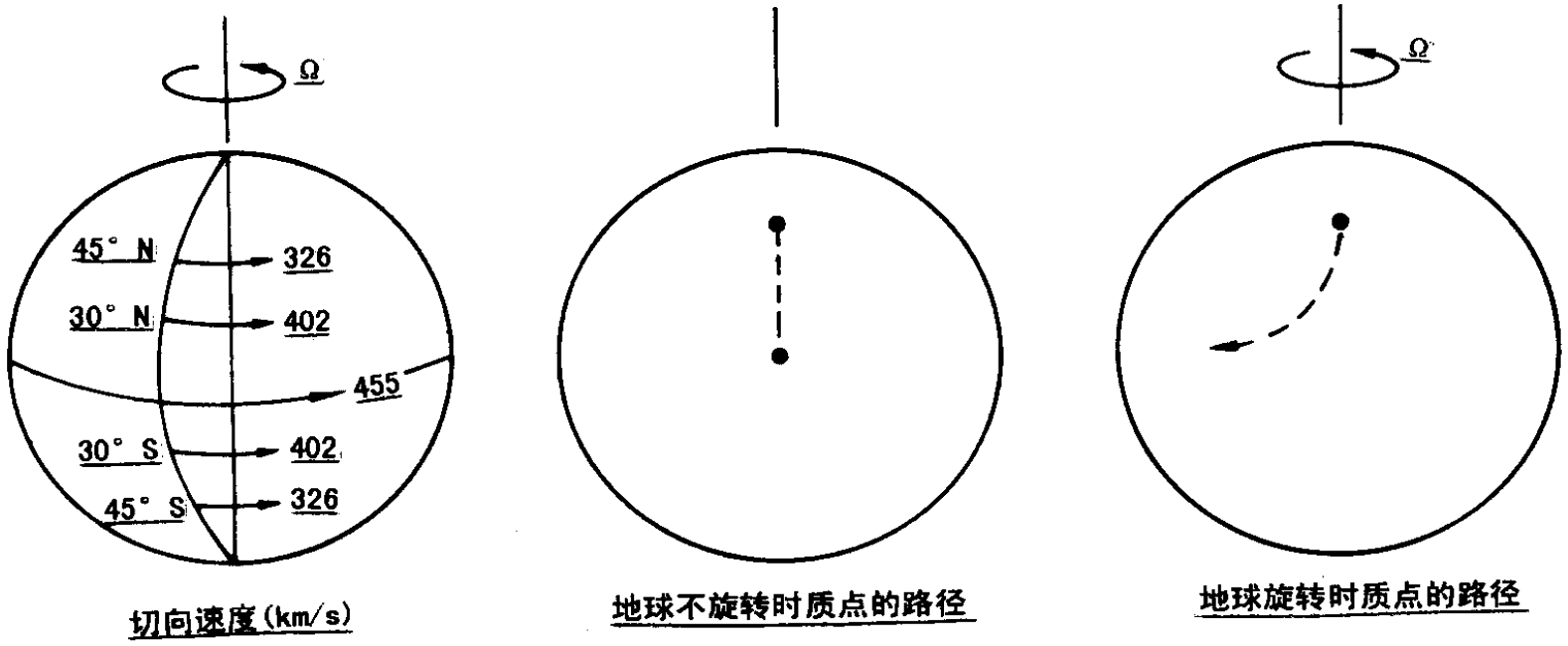
$$-\frac{1}{\rho}\frac{\partial p}{\partial x} = \frac{1}{\rho}\frac{p_B - p_A}{\Delta x} = g\frac{\Delta z}{\Delta x} = g\tan\theta$$

科氏力

科氏力并不是真实的力

它是为了考虑地球旋转效应而人为添加的一种表观力

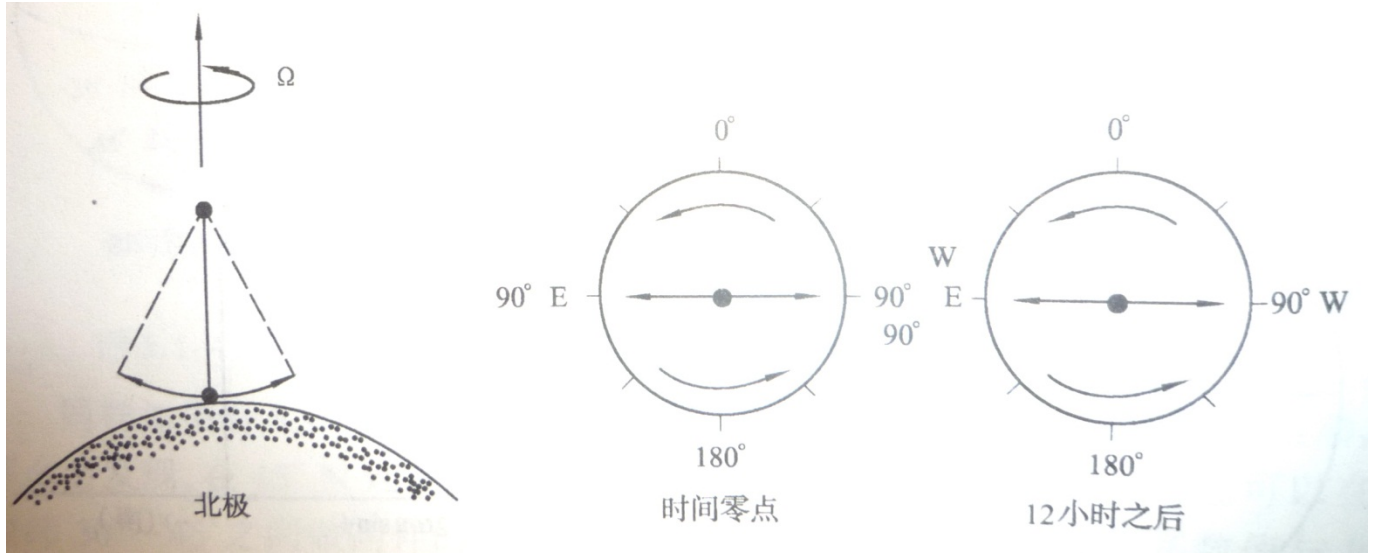
大多数情况下，外力和加速度很大的运动，地球旋转效应可以忽略



在地球上沿南北向运动的质点

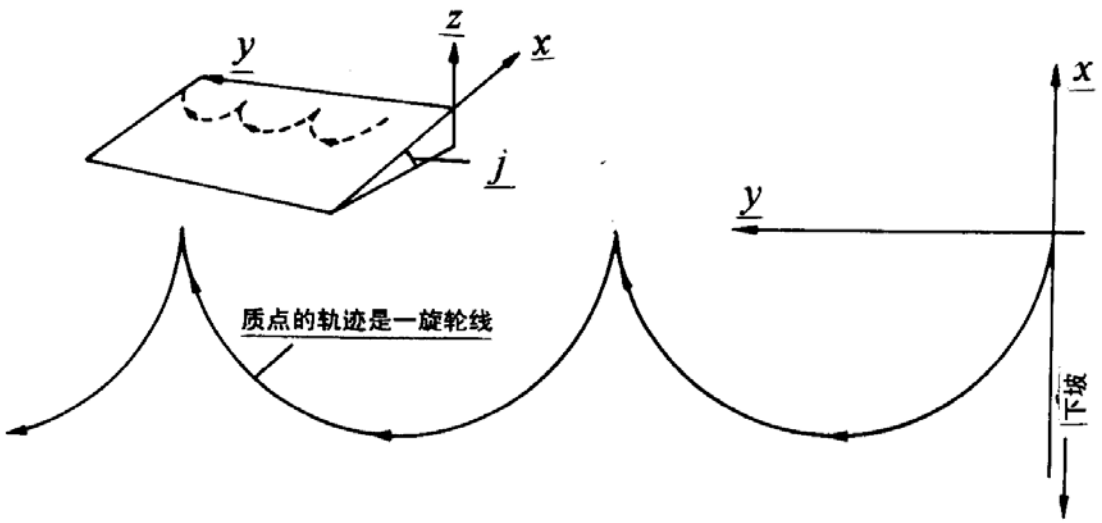
科氏力

傅科摆



傅科摆在南半球运动

无摩擦斜面运动



科氏力

The Coriolis Term in the Momentum Equation The Coriolis term exists because we describe currents in a reference frame fixed on earth. The derivation of the Coriolis terms is not easy. Henry Stommel, the noted oceanographer at the Woods Hole Oceanographic Institution even wrote a book on the subject with Dennis Moore (Stommel & Moore, 1989).

Usually, we just state that the force per unit mass, the acceleration of a parcel of fluid in a rotating system, can be written:

$$\mathbf{a}_{fixed} = \left(\frac{D\mathbf{v}}{Dt}\right)_{fixed} = \left(\frac{D\mathbf{v}}{Dt}\right)_{rotating} + (2\mathbf{\Omega} \times \mathbf{v}) + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{R}) \qquad (7.14)$$

where $\mathbf{R}$ is the vector distance from the center of earth, $\mathbf{\Omega}$ is the angular velocity vector of earth, and $\mathbf{v}$ is the velocity of the fluid parcel in coordinates fixed to earth. The term $2\mathbf{\Omega} \times \mathbf{v}$ is the Coriolis force, and the term $\mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{R})$ is the centrifugal acceleration. The latter term is included in gravity (figure 7.4).

科氏力

$$\begin{cases} x\text{方向：} fv - 2\Omega w\cos\varphi \\ y\text{方向：} -fu \\ z\text{方向：} 2\Omega u\cos\varphi \end{cases}$$

$$f = 2\Omega\sin\varphi$$

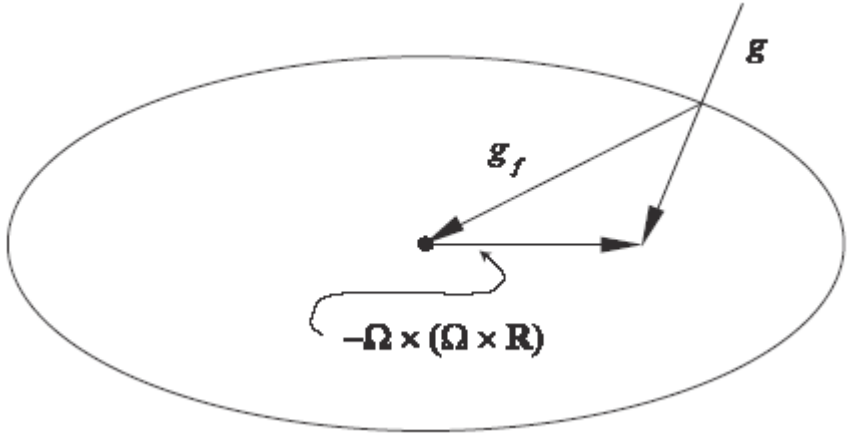
the magnitude Ω of $\mathbf{\Omega}$ is the *Rotation Rate of earth*, 2π radians per sidereal day or

$$\boxed{\Omega = 7.292 \times 10^{-5} \text{ radians/s}} \tag{7.11}$$

科氏力

- 科氏力与质点相对于地球运动的速度成正比，如果质点相对地球静止，科氏力为零
- 科氏力随着纬度的增加而增大，在两极达到最大值，在赤道为零
- 科氏力总是与质点运动方向成直角，在北半球位于运动方向的右侧，南半球位于左侧

重力



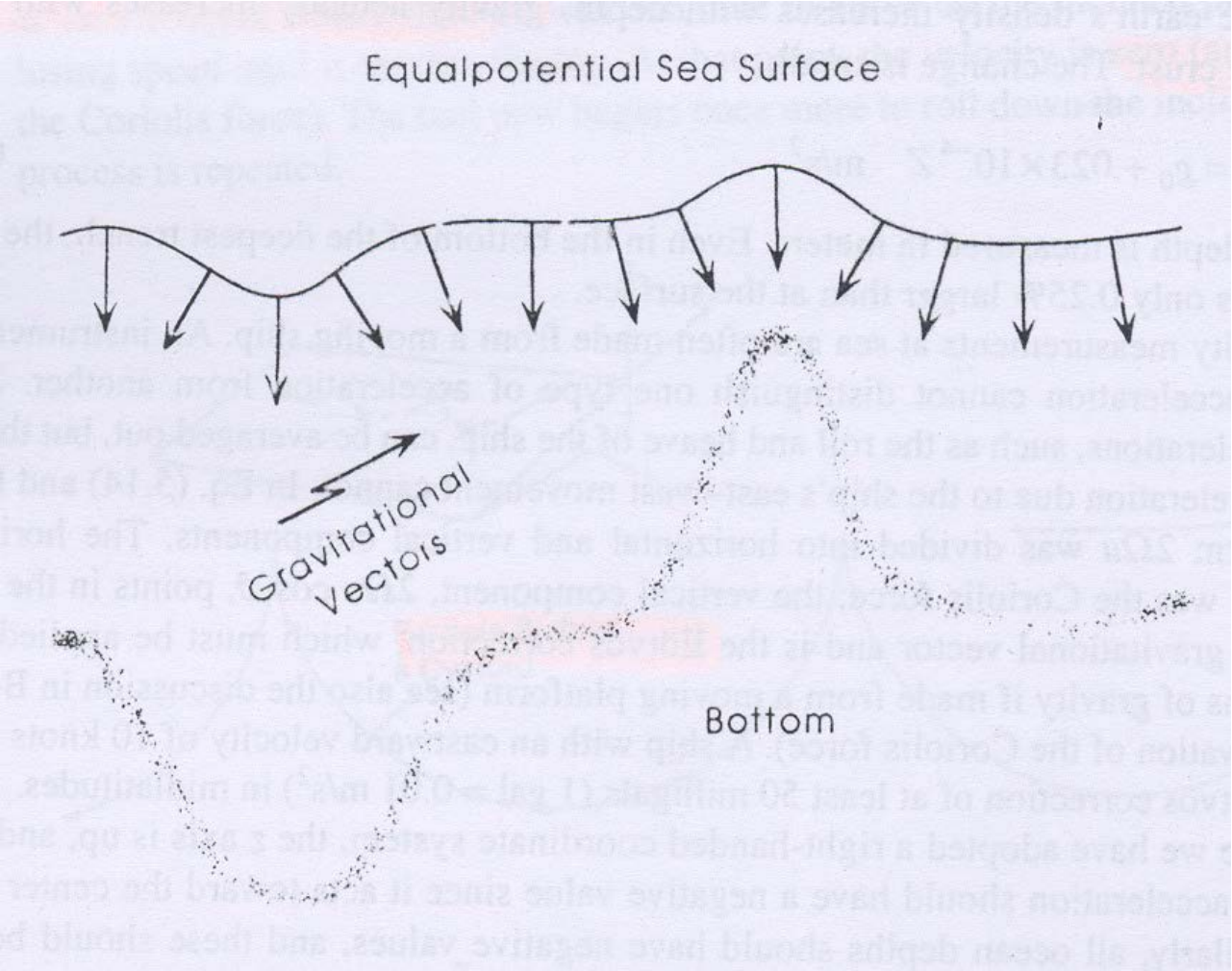
$$\mathbf{g} = \mathbf{g}_f - \Omega \times (\Omega \times \mathbf{R})$$

$$\mathbf{g}_f = \frac{G\,M_E}{R^2}$$

重力加速度在赤道地区为9.78m/s^2，在极地为9.83m/s^2，随纬度增加而增加

这主要是由于向心加速度的变化和地球半径的变化

重力



- 与重力矢相垂直的面称为**等势面**，质点沿等势面运动时重力势能不变
- **平均状态下的海平面**就可以近似看作是等势面

Navier-Stokes方程、Euler 方程

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = -\,\frac{1}{\rho}\,\frac{\partial p}{\partial x} + 2\,\Omega\,v\,\sin\varphi + F_x$$

$$\frac{\partial v}{\partial t} + u\,\frac{\partial v}{\partial x} + v\,\frac{\partial v}{\partial y} + w\,\frac{\partial v}{\partial z} = -\,\frac{1}{\rho}\,\frac{\partial p}{\partial y} - 2\,\Omega\,u\,\sin\varphi + F_y$$

$$\frac{\partial w}{\partial t} + u\,\frac{\partial w}{\partial x} + v\,\frac{\partial w}{\partial y} + w\,\frac{\partial w}{\partial z} = -\,\frac{1}{\rho}\,\frac{\partial p}{\partial z} + 2\,\Omega\,u\,\cos\varphi - g + F_z$$

边界条件 Boundary Conditions

In fluid mechanics, we generally assume:

- No velocity normal to a boundary, which means there is no flow through the boundary
  “无穿透” 条件：流体在边界处的法向速度为零，意味着流体不能穿透边界
- No flow parallel to a solid boundary, which means no slip at the solid boundary
  “无滑移” 条件：流体在固体边界处的切向速度为零

无穿透条件：确保流体不会穿过固体边界，保证质量守恒，符合实际物理现象
无滑移条件：流体分子与固体表面相互作用，最靠近壁面的流体层粘附在壁面上，是粘性流体的重要特征
应用场景：管道流动分析、翼型绕流计算、边界层理论、计算流体动力学(CFD)模拟
这些假设是流体力学建模的基础,在大多数工程应用中都能得到很好的验证

静压假定 Hydrostatic Approximation

静压近似是指对水平尺度远大于铅直尺度的大尺度运动

垂向动量守恒方程可以简化为静压关系

海洋中大多数的运动基本满足静压假定

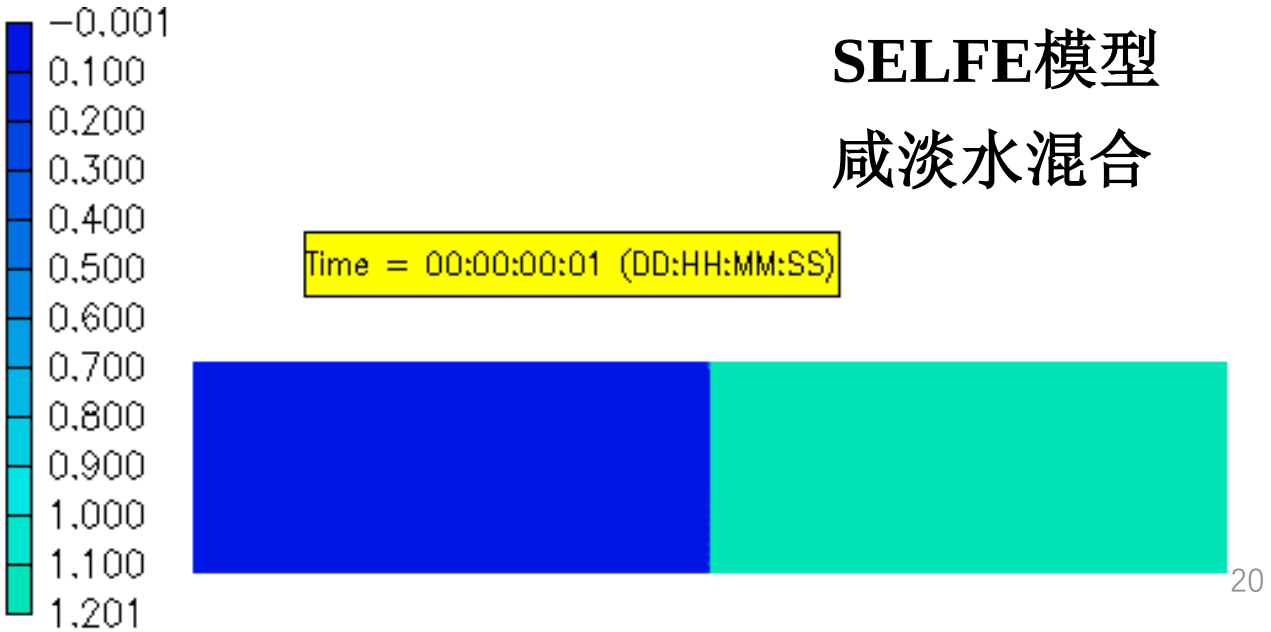
但研究小尺度、垂向运动相对较大时，静压假定将不满足，此时应使用非静压假定

一个非静压假定的例子:

两种不同盐度的水体混合运动

SELFE model

FVCOM model



SELFE模型

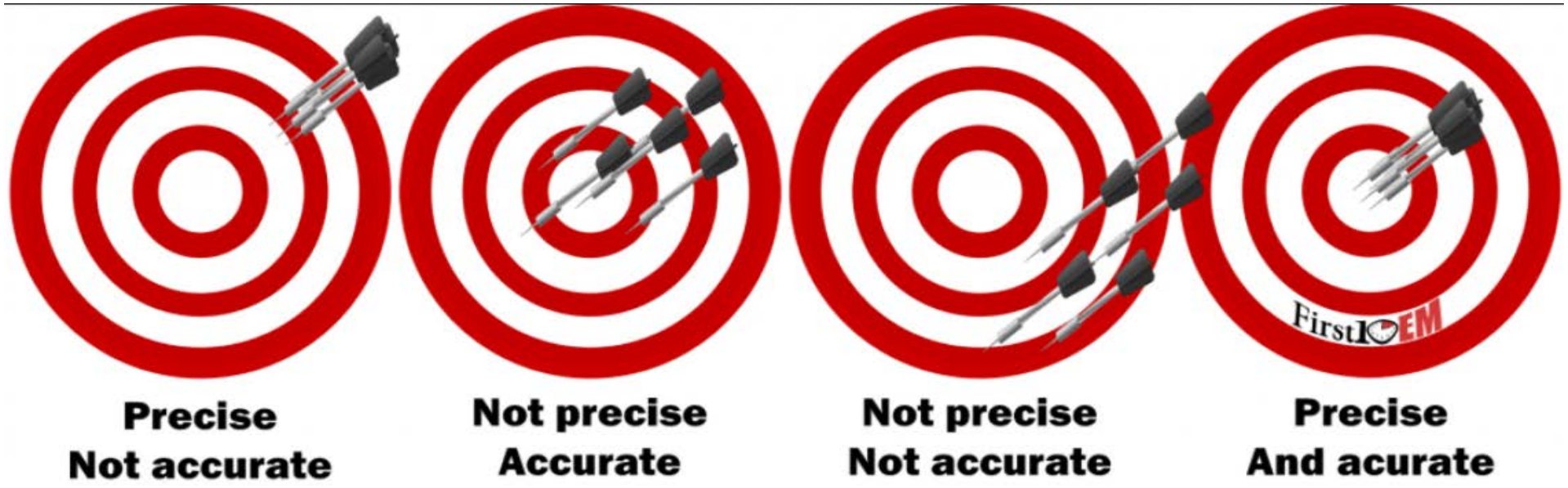
咸淡水混合

长江口的动量平衡分析

即分别计算动量方程中的各分项，并比较其大小

长江口潮差	潮波长度	纬度	水流平均流速	风速
2.7	400	31	0.6	10
m	km	北纬，度	m/s	m/s
			东向	东风
			南北向流速为0	无其他风向
空气密度	**海水密度**	**Ω**	**平均水深**	**近底部流速**
1.237	1025	7.29E-05	10	0.3
kg/m^3	kg/m^3	弧度/s	m	m/s

Precise = 精确，　Accurate = 准确



- 动量方程（$F = ma$）：

$$\begin{cases} \frac{du}{dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + 2\Omega v\sin\varphi + A_h\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + A_z\frac{\partial^2 u}{\partial z^2} \\ \frac{dv}{dt} = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial y} - 2\Omega u\sin\varphi + A_h\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + A_z\frac{\partial^2 v}{\partial z^2} \\ \frac{dw}{dt} = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + 2\Omega u\cos\varphi - g + A_h\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\right) + A_z\frac{\partial^2 w}{\partial z^2} \end{cases}$$

等号左边： 加速度 = 局地加速度项 + 平流加速度项（平流项、非线性项）

等号右边： 压强梯度力项 + 科氏力项 + 水平湍流粘性项 + 垂直湍流粘性项 + 重力项

- 静压假定，$\mathrm{d}w / \mathrm{d}t = 0$，即重力与浮力平衡：

$$-\frac{1}{\rho}\frac{\partial p}{\partial z} = g$$

- 连续方程(质量守恒)：

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

- 假设流体不可压缩，即密度ρ为常数，则连续方程简化为：

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$